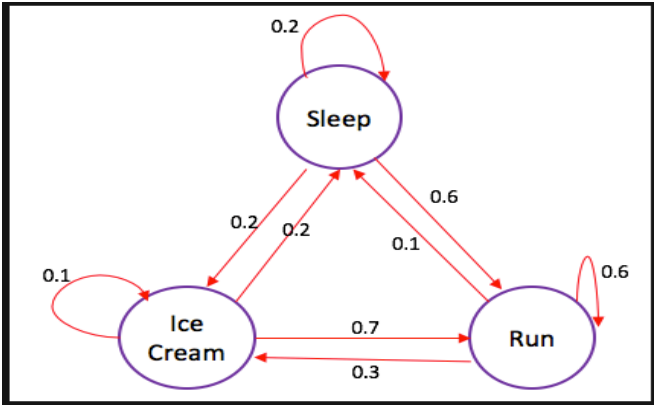
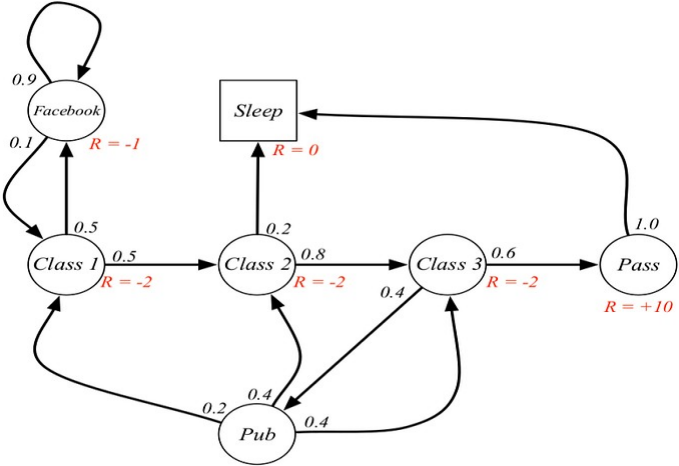


1	Define a Support Vector Machine (SVM) and explain its primary purpose in machine learning.															
2	Define the k-Nearest Neighbours (KNN) algorithm and explain its fundamental principle.															
3	Define the CART algorithm and explain its primary purpose in machine learning															
4	Define Bagging as ensemble methods in machine learning															
5	State the issues in decision tree learning															
6	Justify the statement “KNN is called as lazy learner algorithm”															
7	Define Boosting as ensemble methods in machine learning															
8	Explain the core idea behind Bagging and how it differs from Boosting in terms of how base models are combined.															
9	Describe the concept of a margin in SVMs. How does the margin relate to the separation of data points in SVM classification?															
10	Compare and contrast linear and non-linear SVMs. Discuss when you would choose one over the other															
11	Discuss the impact of choosing a larger value of "k" in KNN. How does it affect the model's bias and variance, and what considerations should be made when selecting an appropriate "k"?															
12	Consider the data with attributes (acid durability and strength) to classify whether a special paper tissue is good or not. For new sample with $X_1=3$ and $x_2=7$, predict the classification value using KNN algorithm with $K=3$ <table><tr><th>X1-Acid durability</th><th>X2-Strengh</th><th>y-Classification</th></tr><tr><td>7</td><td>7</td><td>Bad</td></tr><tr><td>7</td><td>4</td><td>Bad</td></tr><tr><td>3</td><td>4</td><td>Good</td></tr><tr><td>1</td><td>4</td><td>Good</td></tr></table>	X1-Acid durability	X2-Strengh	y-Classification	7	7	Bad	7	4	Bad	3	4	Good	1	4	Good
X1-Acid durability	X2-Strengh	y-Classification														
7	7	Bad														
7	4	Bad														
3	4	Good														
1	4	Good														
13	Compare and contrast the Gini impurity and entropy as criteria for splitting nodes in a decision tree.															

14	For the given data of sports, on which age and gender are taking a part in a decision on ‘what kind of person would play ground-game? Find the root node of a decision tree using CART. Note: consider age ≤ 25 and age > 25. <table><tr><th>Age</th><th>Gender</th><th>Sportive</th></tr><tr><td>22</td><td>F</td><td>yes</td></tr><tr><td>24</td><td>M</td><td>yes</td></tr><tr><td>30</td><td>F</td><td>yes</td></tr><tr><td>31</td><td>F</td><td>no</td></tr><tr><td>27</td><td>F</td><td>no</td></tr><tr><td>32</td><td>M</td><td>no</td></tr><tr><td>25</td><td>F</td><td>yes</td></tr><tr><td>30</td><td>M</td><td>no</td></tr><tr><td>24</td><td>F</td><td>yes</td></tr><tr><td>21</td><td>F</td><td>yes</td></tr><tr><td>29</td><td>M</td><td>yes</td></tr><tr><td>26</td><td>M</td><td>no</td></tr><tr><td>21</td><td>M</td><td>no</td></tr></table>	Age	Gender	Sportive	22	F	yes	24	M	yes	30	F	yes	31	F	no	27	F	no	32	M	no	25	F	yes	30	M	no	24	F	yes	21	F	yes	29	M	yes	26	M	no	21	M	no																																																
Age	Gender	Sportive																																																																																									
22	F	yes																																																																																									
24	M	yes																																																																																									
30	F	yes																																																																																									
31	F	no																																																																																									
27	F	no																																																																																									
32	M	no																																																																																									
25	F	yes																																																																																									
30	M	no																																																																																									
24	F	yes																																																																																									
21	F	yes																																																																																									
29	M	yes																																																																																									
26	M	no																																																																																									
21	M	no																																																																																									
15	For the given dataset, draw a decision tree using Gini index <i>PlayTennis: training examples</i> <table><tr><th>Day</th><th>Outlook</th><th>Temperature</th><th>Humidity</th><th>Wind</th><th>PlayTennis</th></tr><tr><td>D1</td><td>Sunny</td><td>Hot</td><td>High</td><td>Weak</td><td>No</td></tr><tr><td>D2</td><td>Sunny</td><td>Hot</td><td>High</td><td>Strong</td><td>No</td></tr><tr><td>D3</td><td>Overcast</td><td>Hot</td><td>High</td><td>Weak</td><td>Yes</td></tr><tr><td>D4</td><td>Rain</td><td>Mild</td><td>High</td><td>Weak</td><td>Yes</td></tr><tr><td>D5</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Weak</td><td>Yes</td></tr><tr><td>D6</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Strong</td><td>No</td></tr><tr><td>D7</td><td>Overcast</td><td>Cool</td><td>Normal</td><td>Strong</td><td>Yes</td></tr><tr><td>D8</td><td>Sunny</td><td>Mild</td><td>High</td><td>Weak</td><td>No</td></tr><tr><td>D9</td><td>Sunny</td><td>Cool</td><td>Normal</td><td>Weak</td><td>Yes</td></tr><tr><td>D10</td><td>Rain</td><td>Mild</td><td>Normal</td><td>Weak</td><td>Yes</td></tr><tr><td>D11</td><td>Sunny</td><td>Mild</td><td>Normal</td><td>Strong</td><td>Yes</td></tr><tr><td>D12</td><td>Overcast</td><td>Mild</td><td>High</td><td>Strong</td><td>Yes</td></tr><tr><td>D13</td><td>Overcast</td><td>Hot</td><td>Normal</td><td>Weak</td><td>Yes</td></tr><tr><td>D14</td><td>Rain</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr></table>	Day	Outlook	Temperature	Humidity	Wind	PlayTennis	D1	Sunny	Hot	High	Weak	No	D2	Sunny	Hot	High	Strong	No	D3	Overcast	Hot	High	Weak	Yes	D4	Rain	Mild	High	Weak	Yes	D5	Rain	Cool	Normal	Weak	Yes	D6	Rain	Cool	Normal	Strong	No	D7	Overcast	Cool	Normal	Strong	Yes	D8	Sunny	Mild	High	Weak	No	D9	Sunny	Cool	Normal	Weak	Yes	D10	Rain	Mild	Normal	Weak	Yes	D11	Sunny	Mild	Normal	Strong	Yes	D12	Overcast	Mild	High	Strong	Yes	D13	Overcast	Hot	Normal	Weak	Yes	D14	Rain	Mild	High	Strong	No
Day	Outlook	Temperature	Humidity	Wind	PlayTennis																																																																																						
D1	Sunny	Hot	High	Weak	No																																																																																						
D2	Sunny	Hot	High	Strong	No																																																																																						
D3	Overcast	Hot	High	Weak	Yes																																																																																						
D4	Rain	Mild	High	Weak	Yes																																																																																						
D5	Rain	Cool	Normal	Weak	Yes																																																																																						
D6	Rain	Cool	Normal	Strong	No																																																																																						
D7	Overcast	Cool	Normal	Strong	Yes																																																																																						
D8	Sunny	Mild	High	Weak	No																																																																																						
D9	Sunny	Cool	Normal	Weak	Yes																																																																																						
D10	Rain	Mild	Normal	Weak	Yes																																																																																						
D11	Sunny	Mild	Normal	Strong	Yes																																																																																						
D12	Overcast	Mild	High	Strong	Yes																																																																																						
D13	Overcast	Hot	Normal	Weak	Yes																																																																																						
D14	Rain	Mild	High	Strong	No																																																																																						
16	Discuss the theoretical foundations and practical applications of Support Vector Machines (SVM) in machine learning. Illustrate your answer with appropriate examples and diagrams where necessary.																																																																																										
17	Compare and contrast Bagging and Boosting in terms of their approach to handling model diversity. How does each method encourage diversity among base models, and how does this impact ensemble performance?																																																																																										
18	Compare and contrast the strengths and weaknesses of classification algorithms like KNN, decision trees, and support vector machines. Under what circumstances would you prefer to use these alternatives?																																																																																										

19	Design a scenario in which KNN can be applied to solve a practical problem. Specify the dataset characteristics, explain how you would preprocess the data, and outline the steps to choose the most suitable "k" value for the problem.
20	Design a scenario where an SVM can be applied to solve a practical problem. Outline the dataset requirements, preprocessing steps, kernel choice, and hyperparameter tuning process
21	Suppose you have trained an SVM model for a classification task. Discuss the performance evaluation metrics you would use to assess the model's quality and provide guidelines on interpreting the results.
22	What is the difference between training error and validation error in model selection?
23	Provide a brief overview of why optimization is crucial in the context of training machine learning models.
24	What challenges can arise when applying optimization techniques to large-scale machine learning models?
25	What is the bias-variance tradeoff, and how does it impact model selection?
26	What is the validation set used for in hyperparameter tuning?
27	How does cross-validation help in assessing the performance of a machine learning model?
28	Differentiate between model selection and Model evaluation
29	What are the advantages and disadvantages of using grid search for hyper-parameter tuning?
30	Discuss how Cross-Validation and Regularization techniques be employed to address the issues of overfitting and underfitting?
31	Differentiate between local and global optimization strategies. How do these approaches impact the performance of machine learning algorithms?
32	Compare and contrast stochastic gradient descent (SGD) with mini-batch gradient descent.
33	Clarify the importance of model selection in machine learning. Provide three key factors that influence the decision-making process when selecting a model for a specific task.
34	Explain how gradient descent algorithm works and provide an example of a scenario where it can be applied.
35	Explain the concept of hyper parameter tuning in machine learning. Why is it important, and how does it differ from model training?
36	Differentiate between types of gradient descent algorithm with an example.
37	Describe the concept of cross-validation in machine learning. Explain how k-fold cross-validation works, and highlight its significance in assessing and improving the performance of machine learning models.
38	Describe the algorithm of Steepest Descent method in detail by considering one example.

39	Discuss the importance of model performance metrics in evaluating different hyperparameter settings. Evaluate how Random Search can be a more efficient alternative to Grid Search, including its advantages and potential drawbacks.
40	Discuss the steps taken to select a machine learning model for a classification problem. Include in your response the functions of model training, validation, and testing. How does each step help to ensure that the selected model performs effectively on new data?
41	Discuss the steps for defining the search space, assess performance, and for selecting the best hyperparameters while designing a Random Search-based hypertuning strategy.
42	Discuss, how the learning rate impacts the convergence of Gradient Descent. Discuss the differences between Adaptive Gradient Descent methods (such as AdaGrad, RMSprop, and Adam) and traditional Gradient Descent. Include the advantages and limitations of these adaptive methods.
43	Define semi-supervised learning in the context of machine learning.
44	How does semi-supervised learning leverage both labeled and unlabeled data in training?
45	Define reinforcement learning.
46	Analyze the potential risks or challenges of using semi-supervised learning in situations with limited labeled data.
47	State and explain the advantages of semi-supervised learning
48	Provide examples of real-world applications where reinforcement learning has been successfully employed.
49	Describe two key benefits of using reinforcement in machine learning applications.
50	Discuss the challenges of semi-supervised learning in machine learning
51	Provide an example of a real-world problem where semi-supervised learning would be beneficial.
52	Analyze the benefits of using semi-supervised learning techniques, especially in scenarios with limited labeled data. How does leveraging unlabeled data enhance the accuracy and efficiency of machine learning models?
53	Discuss the application of Autonomous Vehicles and how to improve Decision-making processes for self-driving cars.
54	Explain the key elements of reinforcement learning, including agents, environments, actions, and rewards. How do these elements interact within the reinforcement learning framework?

55	Justify with following diagram that Markov Process is the memory less random process 
56	For the given Markov Decision proces calculate rewards for a) Start state is Class 1, move to Class 2, then class 3, then pass, then sleep. b) Start at pub and go to sleep through class 2 . 
57	What is Reinforcement Learning? How does it differ from other types of Machine Learning, such as Supervised Learning and unsupervised learning?
58	How is Reinforcement Learning utilized in game AI? Provide an example of a popular game where Reinforcement Learning has been applied.
59	Explain the concept of the reward function in Reinforcement Learning. Why is it important for the learning process? Evaluate the limitations and challenges faced by reinforcement learning methods. Propose potential solutions
60	Explain the basic idea behind Q-learning as a Reinforcement Learning algorithm. How does it help an agent learn optimal actions?
61	Elaborate the application of Reinforcement Learning in robotics. How is it used to improve robotic behaviors and optimize decision-making?

62	Describe the benefits of using semi-supervised learning techniques, especially in scenarios with limited labeled data. How does leveraging unlabeled data enhance the accuracy and efficiency of machine learning models?
63	Analyze specific use cases of reinforcement learning in different domains, such as robotics, gaming, or autonomous systems. How does reinforcement learning optimize decision-making in these applications?

4	5
4	10
4	10
4	10
4	10

4	10
4	10
4	10
5	2
5	2
5	2
5	2
5	2
5	2
5	2
5	2
5	5
5	5
5	5
5	5
5	5
5	5
5	5
5	10
5	10
5	10

5	10
5	10
5	10
5	10
6	2
6	2
6	2
6	2
6	2
6	2
6	2
6	5
6	5
6	5
6	5
6	5

6	5
6	5
6	10
6	10
6	10
6	10
6	10

6	10
6	10

Sr No.	Questions	CO	RBT
1	Define semi-supervised learning in the context of machine learning.	6	U
2	How does semi-supervised learning leverage both labeled and unlabeled data in training?	6	AN
3	Analyze the potential risks or challenges of using semi-supervised learning in situations with limited labeled data.	6	R
4	State and explain the advantages of semi-supervised learning.	6	AN
5	Provide examples of real-world applications where semi-supervised learning is used.	6	U
6	Describe two key benefits of using reinforcement learning.	6	A
7	Discuss the challenges of semi-supervised learning.	6	U
8	Provide an example of a real-world problem where semi-supervised learning is applied.	6	AN
9	Explain the application of Autonomous Learning in machine learning.	6	A
10	How does leveraging unlabeled data enhance the learning process for self-driving cars?	6	AN
11	Discuss the application of Autonomous Learning, including its benefits and challenges.	6	C
12	Explain the role of reinforcement learning in autonomous systems.	6	AN
13	Describe how reinforcement learning is used in robotics.	6	AN
14	Explain the concept of Markov Decision Process (MDP) in reinforcement learning.	6	AN
15	How does a Markov Decision Process differ from other types of Machine Learning?	6	AN
16	Provide an example of a popular game where Reinforcement Learning is applied.	6	A
17	Explain the basic idea behind Q-learning as a Reinforcement Learning algorithm.	6	E
18	Evaluate the limitations and challenges of learning in robotics.	6	E
19	How is it used to improve robotic behaviors and optimize performance?	6	C
20	Discuss the application of Autonomous Learning in different domains, such as gaming or autonomous systems.	6	E
21	How does reinforcement learning optimize unlabeled data enhance the accuracy and performance of autonomous systems.	6	AN

How does reinforcement learning optimize

PI	Marks	Type
		theoretical or numerical
2.5.1	2	
4.4.2	2	T
1.6.2	2	T
2.5.2	2	T
3.5.1	2	T
3.5.1	2	T
2.6.4	2	T
2.5.2	5	T
3.5.1	5	T
2.6.4	5	T
4.6.3	5	T
2.6.4	5	T
2.6.4	5	N
2.6.4	5	N
4.6.3	10	T
3.5.1	10	T
4.4.2	10	T
4.4.2	10	T
2.5.2	10	T
4.4.2	10	T
2.5.2	10	T